

DSV: The Case For Blockchain Payments

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DSV is a leading global transport and logistics company, founded in 1976 in Denmark. They offer air, sea, road, and train freight services across over 90 countries with over 150,000 employees. Given the norms of the logistics industry, it is considered an “asset-light” giant. Considering its recent acquisition of DB Schenker, it is a top global freight forwarder.

The focus of this analysis will centre on DSV’s global payment architecture. The argument will be made that the implementation of blockchain-based settlement rails can virtually eliminate financial friction in the current legacy system. More specifically, by transitioning the cross-border treasury operations from legacy correspondent banking to stablecoin-denominated atomic settlement, DSV can increase its EBITDA margins through i) reduction of FX spreads, ii) mitigation of counterparty risk, and iii) compression of the Cash Conversion Cycle (CCC). The analysis will be capped with a risk-assessment, evaluating the degree of certainty with which blockchain transformation can increase the business’ performance, including data sources, the overall fintech partnership landscape, and management considerations. Ultimately, it will be shown that blockchain payments can significantly transform the global logistics industry, especially its characteristic multi-day liquidity gaps and costly reconciliation.

I. BUSINESS & PAYMENTS REALITY

A. Business Overview

DSV is a global orchestrator of transport and logistics. The firm operates an "asset-light": business model. Unlike “asset-heavy” carriers like (Maersk or FedEx) that own the ships and planes, DSV owns the data, the contracts, and the customer relationships. They act as a high-volume broker, purchasing “belly space” on aircraft and container slots on ships at wholesale rates and reselling them to shippers. This model allows for extreme financial flexibility. When global trade volumes dip, DSV is not burdened by the fixed costs of maintaining an idle fleet.

DSV is quite robust in that it does not rely on a single industry. Its portfolio is diversified across time-sensitive verticals. It employs a "barbell" segmentation strategy designed to balance massive operational volumes with high-yield profitability. The first tier consists of Global Accounts (multinational corporations such as Apple, BMW, and Walmart) which typically account for approximately 30% of net revenue. These customers provide the high-volume “baseload” that allows DSV to negotiate preferential wholesale rates with carriers. While these contracts are lower-margin due to the immense bargaining power of the clients, they are strategically “sticky,” often involving multi-year service agreements and deep integration into the customer's supply chain through dedicated “Control Tower” teams.

The second tier comprises a long tail of thousands of Small and Medium Enterprises (SMEs). These customers lack the scale to negotiate directly with global shipping lines and therefore pay higher retail spreads, driving a significantly higher Gross Profit per unit (Yield) for DSV. By late 2025, DSV reported a market-leading conversion ratio of approximately 29.3%, largely sustained by this high-margin SME segment.

B. Payments Flow

Revenue Streams

Post-acquisition of DB Schenker in 2025, the firm’s scale has effectively doubled. DSV’s revenue is generated through three core divisions. These three divisions are, i) Air & Sea, ii) Road, and iii) Solutions (Contract Logistics).

Air & Sea is the highest margin-division. DSV manages the complex documentation, customs brokerage and port-to-port movement of goods. Profit is derived from the “spread” between what they pay the carrier and what they charge the customer.

Revenue	DSV buys “slots” on ships and planes at wholesale prices. They then sell those slots to customers (like Apple or a small toy company) at retail prices.
The Gross Profit (Yield)	The difference between wholesale price and retail price is called the “Yield”. E.g. DSV buys a container slot for \$1,500, sells for \$2,000 (Yield is \$500).

The Road Division includes a vast European and North American trucking network using a mix of owned and subcontracted drivers to move cargo across continents. The profit driver is network density. Profit increases when DSV can ensure trucks are never driving empty or “backhauling”. Optimization of routes is also critical for profit.

Revenue	Fees are charged to customers for point-to-point ground transportation.
The Gross Profit (Yield)	This is the difference between the customer’s payment and the cost of the subcontractor (the truck driver) or the direct fuel and maintenance costs.

Solutions or the Contract Logistics Division involves managing customer warehouses and inventory. It creates “sticker” revenue, as customers are less likely to switch providers once their inventory is integrated into DSV’s facility. The profit driver is automation. Using robotics, or efficient software (e.g. GenAI) to pick items faster with fewer people allows DSV to keep more of the service fee as a profit.

Revenue	Generated through long-term service contracts. DSV charges for Storage (renting shelf space) and Handling (fees for items picked/packed/shipped).
The Gross Profit (Yield)	The total contract fees minus the cost of warehouse labour, rent, and utilities.

Importantly, the Conversion Ratio is used by DSV to measure efficiency of each division:

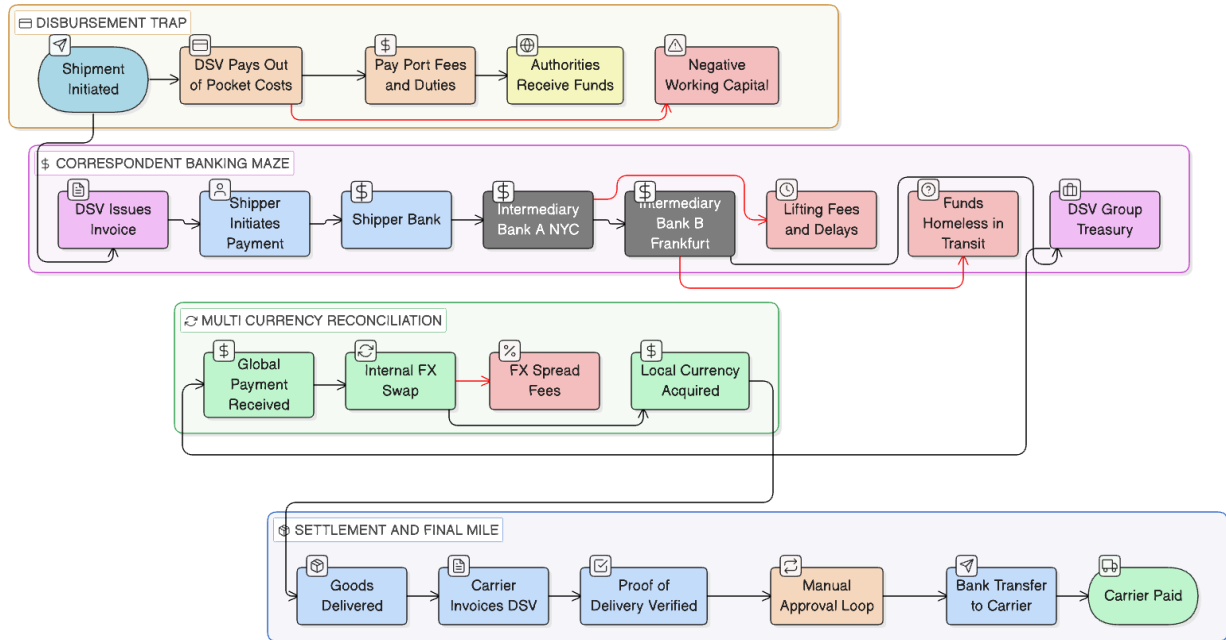
$$\text{Conversion Ratio} = \frac{\text{EBIT (Operating Profit)}}{\text{Gross Profit}}$$

DSV’s goal is always to keep this ratio high (approximately 30%).

Payment Phases Master Diagram:

To better understand the payments flow, this chart was created to illustrate the primary phases of fund-transfers that result in DSV being paid and the carriers being paid.

The payment lifecycle is divided into four phases to reflect the chronological and jurisdictional reality of moving goods across borders. Phase I (Disbursement) represents the initial cash outlay required to unlock cargo from ports, forcing DSV to pre-fund taxes before receiving revenue. Phase II (Correspondent Banking) covers the inbound revenue path, where a customer’s payment must hop through multiple intermediary banks to reach DSV HQ. Phase III (Multi-Currency Reconciliation) is the treasury function where global revenue is converted into local currencies to fund operations. Finally, Phase IV (Settlement) is the end of the cycle, where DSV pays the actual transporters (carriers) after manually verifying that the job was completed.



Note that in the diagram created above, the black arrows represent a normal payment flow, while the red arrows represent friction points (fee leakage) - in other words, inefficient payment processes.

While all four phases contain friction, the current legacy model is most inefficient in Phase II (Correspondent Banking) and Phase III (Multi-Currency Reconciliation). These stages suffer from dead time (3–5 day settlement delays). Blockchain infrastructure can be implemented across both of these phases to transition from batch-based bank messaging. Additionally, Phase IV (Settlement) can implement blockchain to automate the manual verification process - especially to ensure carriers are paid the moment a digital “Proof of Delivery” is logged (this is an idiosyncrasy of the trade industry).

C. Key Constraints

Economically, DSV is constrained by the working capital gap. They must pre-fund port disbursements (Phase I) while waiting 3-5 days for inbound SWIFT settlements (Phase II), they carry a massive “float” that incurs significant opportunity costs and interest expenses. Operationally, the system is tethered to manual reconciliation. The reliance on paper-based Proof of Delivery (POD) and human approval loops in Phase IV makes real-time settlement impossible under legacy banking rails. Regulatorily, DSV must navigate a fragmented landscape of more than 90 jurisdictions, each with distinct AML/KYC requirements and capital controls, which forces the use of the multi-hop correspondent banking network to ensure compliance. Finally, the system faces inherent Risk-Related constraints, specifically of a counterparty and FX nature. In the time it takes for funds to move through the banking maze, currency fluctuations can erode the thin margins of a freight contract. Moreover, the lack of visibility into “homeless funds” in transit creates a lack of financial certainty that hampers precise treasury forecasting.

D. The Significance of Payments

For an asset-light orchestrator like DSV, payments are not back-office functions. They are the primary determinant of profitability and operational scaling. In a high-volume industry where net margins are historically thin (DSV reported a conversion ratio of 29.3% in late 2025) even marginal “leakage” in the form of FX spreads (often 1-3%) and correspondent bank fees can represent a significant percentage of EBIT. Furthermore, payments directly impact growth through the Cash Conversion Cycle (CCC) - every day that capital is trapped in the “Correspondent Maze” or used to pre-fund disbursements is a day that capital cannot be deployed for M&A, such as the \$15.9 billion acquisition of DB Schenker (Reuters, 2024). Finally, payments are the bedrock of operational risk. According to a report by Juniper Research, B2B cross-border payment friction costs businesses billions annually in manual reconciliation and delayed settlements, which in DSV’s case, can lead to carrier blacklisting and supply chain paralysis if payouts are not timely.

II. BLOCKCHAIN TRANSFORMATION HYPOTHESIS

A. The Unified Settlement Layer

The proposed transformation replaces the fragmented correspondent banking maze with a unified, blockchain-based settlement rail. Instead of relying on the SWIFT messaging network, DSV would implement a Stablecoin Treasury System (using regulated assets like USDC or EURC).

- Currently, DSV pays port authorities (disbursements) days before receiving funds from customers. In the transformed flow, a Smart Contract replaces the manual invoice. The customer funds the contract upon booking. The moment the “Bill of Lading” is digitally verified, the contract atomically splits the funds, paying the port authorities, the carrier, and DSV’s margin simultaneously.
- This would move accounts from “Nostro/Vostro” to “Peer-to-Peer”: The chain of intermediary banks (Bank A → Bank B → Bank C) is removed. Payment moves directly from the Shipper’s Wallet to DSV’s Corporate Wallet.

B. The Paradigm-Shift of Blockchain

Legacy fintech (like Stripe or standard wire transfers) can digitize the interface, but they cannot fix the settlement rails. Blockchain infrastructure enables three capabilities that legacy systems cannot replicate:

1. Programmable Escrow (Trust-less Settlement)

Legacy systems require credit insurance or Letters of Credit because parties do not trust each other. A smart contract eliminates this counterparty risk by holding funds in escrow and releasing them only when cryptographic proof of delivery is received.

2. 24/7/365 Liquidity

Global trade does not stop on weekends - but banks do. Blockchain enables DSV to settle “Cash-on-Glass” payments to carriers in emerging markets on a Sunday, preventing goods from incurring demurrage (storage) fees at ports.

3. Composite Atomic Transactions

A single blockchain transaction can execute multiple outputs. For example, paying the tax authority in Singapore, the trucker in Vietnam, and the insurer in London instantly in the same block. Legacy rails require three separate, asynchronous wire transfers which is clearly inferior in terms of efficiency.

C. Scope of Change

To ensure this proposal is operationally realistic, it is not proposed to move the entire business on-chain. The suggestion is a back-end transformation and should clarify some points:

- The ERP Layer should stay off-chain. DSV’s order management systems (SAP/Oracle), customer booking portals (myDSV), and legal contracting remain on legacy servers. Customers should still

pay via bank transfer if they choose, with a “Fiat-to-Crypto” gateway partner (like Circle or Stripe Bridge) handling the conversion behind the scenes.

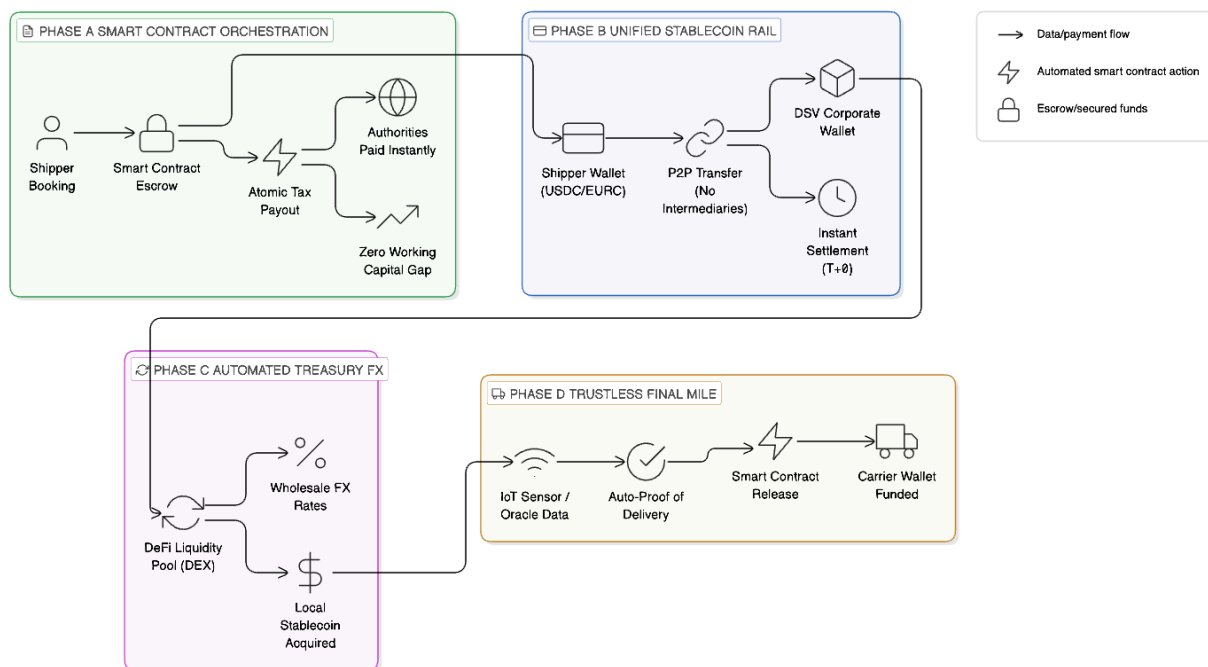
- The Settlement Layer should move on-chain. The movement of high-value liquidity between DSV subsidiaries (Netting), payments to ocean carriers, and the holding of working capital reserves should move to on-chain ledger addresses.
- It is suggested that a bridge be implemented referred to as a “Data Oracle” (e.g. Chainlink) to connect the off-chain environment (e.g. IoT sensors on containers) to the on-chain smart contracts, triggering payments automatically.

D. Magnitude of Impact

The financial impact of this transformation is material and directly accretive to DSV’s bottom line.

- DSV operates with net margins generally under 10%. By replacing the 1.5% - 2.5% cumulative leakage (lifting fees and commercial FX spreads) with a sub-0.1% blockchain network fee and tight stablecoin spreads, DSV could realistically capture \$50M - \$100M in annual incremental EBITDA given their post-acquisition volume.
- Compressing the settlement time dramatically shortens DSV’s Cash Conversion Cycle (CCC). Reducing the float releases hundreds of millions in trapped working capital, which can be redeployed as needed.
- The removal of “aging receivables” via smart contract escrow reduces bad debt expense, which further stabilizes the Conversion Ratio.

Below is a diagram providing an illustration of a blockchain-enhanced workflow for DSV:



III. DILIGENCE & VALIDATION PLAN

To move from a theoretical hypothesis to an investment-grade execution plan, the following diligence workstreams must be completed. This plan is designed to validate the technical feasibility and economic impact of the blockchain transition while addressing the risks inherent in DSV's scale.

A. Assumptions & Uncertainty

The core thesis rests on three critical assumptions:

1. The “Last-Mile” Liquidity Gap

It is assumed that carriers in emerging markets (e.g. Vietnam, Brazil, India) can off-ramp stablecoins to local fiat without prohibitive fees. If local banking restrictions or lack of liquidity in EURC/VND pools create a 2% spread, the efficiency gains are negated.

2. Infrastructure Scalability

It is assumed that blockchain rails can handle DSV's post-acquisition volume. With the DB Schenker integration adding significant scale, the system must process millions of micro-transactions (Phase IV) without gas fee spikes.

3. Disconfirming Data

The case would be disconfirmed if Disbursements (Phase I) are legally required to be paid in local fiat via government-approved legacy portals, preventing the use of automated smart contract escrow.

B. Data Sources for Validation

To prove the magnitude of the impact, the following public and proprietary data can be utilized:

- According to the DSV FY 2025 Results, Net Working Capital (NWC) stood at DKK 1,242 million (a decrease from DKK 9,317 million in 2024). How much of this remaining capital is “trapped” in Phase II (Correspondent Maze) vs. operational requirements must be analyzed.
- DSV's 2025 performance showed sequential improvements. We will benchmark the current Yield (Gross Profit per unit) against the estimated 1.5%–2.5% payment leakage to determine if blockchain can drive the 2026 EBIT outlook toward the upper bound of DKK 25.5 billion.
- Use *Dune Analytics* or *DefiLlama* to assess the depth of EURC and USDC pools in DSV's top 10 trade corridors to ensure "Wholesale FX Rates" (Phase C) are achievable.

C. Infrastructure Diligence (Partner Feasibility)

Vetting potential blockchain partners is a must (e.g., Circle, Chainlink, Stripe Bridge). See below table for specific criteria.

ERP Interoperability	Can the blockchain “Data Oracle” (Phase D) pull real-time data from DSV's SAP S/4HANA system without creating security vulnerabilities?
Compliance (MiCA/Travel Rule)	Does the partner provide automated AML/KYC tagging for Carrier Wallets to satisfy EU regulatory standards?
Gas Fee Predictability	In a high-volume freight environment, Layer 2 (L2) solutions are required with sub-cent transaction costs to ensure Phase IV “Flash Payouts” remain economical.

D. Management Questions

Given 15 minutes with anyone in the organization, the Group CFO or Head of Treasury would be chosen and the focus would be on identifying the unseen costs. The rationale for this is that because the business is so complex with so many moving parts, there may be elements of the payment system that are missed. Every single minute would not be laid out, but sectioned into three stages allowing for a more relaxed conversation. Typically, those lead to more informative discussions.

- Minutes 0-5 would be spent on “Quantification”:
What is the precise “Drag” on the ROCE (Return on Capital Employed) caused by the 4-day settlement float? If you released that capital, would you deploy it into debt reduction or M&A?
- Minutes 5-10 would be spent on “Operational Costs”:
How many FTEs (Full Time Employees) currently sit in Shared Service Centers dedicated solely to manual invoice reconciliation? What is the fully loaded cost of that headcount?
- Minute 10-15 would be spent on “Specific Pains”:
Which specific trade lane (e.g., China-US vs. Intra-Asia) currently has the highest 'Failed Payment' rate, and is that failure typically due to liquidity data mismatch or banking error?

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